Affine Lipschitz projections on nearly Euclidean convex bodies

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Abstract

We construct symmetric convex bodies and scalar Lipschitz functions whose canonical affine-projection norms grow at least as the square root of the dimension. The construction persists at any fixed Banach–Mazur distortion greater than one and gives examples with distortion tending to one while the projection norms diverge.

Repository target. This manuscript concerns AIM problem 898 at revision `5d7e26e8444e`. The precise scope is stated below.

1 Main result and proof

The answer to [1, Question 19] and the scalar formulation in AIM 898 is negative. We construct explicit convex bodies and explicit 1-Lipschitz functions for which the canonical affine projection has Lipschitz constant growing at least as a positive multiple of the square root of the dimension.

Theorem 1.1. *There are origin-symmetric isotropic convex bodies $\widetilde{K}_n \subset \mathbb{R}^{n+1}$ of volume one and real functions $\widetilde{f}_n$ on them such that*

$$\operatorname{Lip}_{\widetilde{K}_n}(\widetilde{f}_n) \le 1, \qquad \liminf_{n\to\infty} \frac{\operatorname{Lip}_{\widetilde{K}_n}(P_{\widetilde{K}_n}\widetilde{f}_n)}{\sqrt{n}} \ge c_*,$$

where the explicit positive constant is

$$c_* = \frac{1}{26\sqrt{2\pi}} \int_{-1}^{1} (1-z^2)^2 e^{-z^2/2}\, dz \approx 0.0152883.$$

Consequently no universal finite projection bound exists.

Proof. We begin before isotropic normalization, defining the affine projection using normalized uniform measure. For $y \in B_2^n$ set

$$h(y) = 1 - \|y\|_2^2, \qquad c(y) = \frac{1}{3}\sum_{i=1}^{n} y_i^3, \qquad K_n = \{(x,y) : y \in B_2^n,\ |x - c(y)| \le h(y)\}.$$

Both $h + c$ and $h - c$ are concave on B_2^n: their Hessians are diagonal with entries $-2 + 2y_i$ and $-2 - 2y_i$, respectively. The upper fiber boundary is therefore concave and the lower boundary convex. Thus K_n is convex; it is compact with nonempty interior, and it is origin-symmetric

because h is even and c is odd. Its projection onto the y-coordinates is B_2^n. In addition, $e_0 = (1, 0, \dots, 0)$ has K_n-norm one.

Define the odd C^1 function

$$\varphi(t) = \begin{cases} t^3/3, & |t| \le 1, \\ t - \frac{2}{3}\operatorname{sgn}(t), & |t| \ge 1. \end{cases} \qquad \text{Thus} \quad \varphi'(t) = \min\{t^2, 1\}.$$

Set

$$f_n(y) = \frac{1}{n}\sum_{i=1}^{n} \varphi(\sqrt{n}\, y_i), \qquad g_n(x, y) = f_n(y).$$

Every partial derivative of f_n has absolute value at most $n^{-1/2}$, so $\|\nabla f_n\|_2 \le 1$ everywhere. Since the y-projection of K_n is B_2^n, we have $\|y\|_2 \le \|(x, y)\|_{K_n}$ for every vector (x, y). It follows that $\operatorname{Lip}_{K_n}(g_n) \le 1$.

We calculate the vertical coefficient of its affine projection. Let (X, Y) be uniform on K_n and put $N = n + 4$. The marginal density of Y is proportional to $h(y)$ on B_2^n, and conditionally on Y, the variable X is uniform on $[c(Y) - h(Y), c(Y) + h(Y)]$. Thus

$$\mathbb{E}(\mathsf{X} \mid Y) = c(Y), \qquad \operatorname{Var}(\mathsf{X} \mid Y) = h(Y)^2/3. \tag{1}$$

The distribution of Y also equals that of the first n coordinates of a uniform point on S^{N-1}, because its marginal density is $(1 - \|y\|_2^2)^{(N-n-2)/2} = h(y)$ up to normalization. The elementary spherical moment formula is

$$\mathbb{E}\prod_i Y_i^{2k_i} = \frac{\prod_i (2k_i - 1)!!}{N(N+2)\cdots(N+2k-2)}, \qquad k = \sum_i k_i. \tag{2}$$

Odd-coordinate moments vanish, and $(-1)!! = 1$. This formula follows directly by polar integration of the corresponding Gaussian monomial.

In particular, $\mathbb{E}(Y_i Y_j) = \delta_{ij}/N$ and $\mathbb{E}(\mathsf{X} Y_i) = 1/[N(N+2)]$. Put

$$\ell(y) = \frac{1}{N+2}\sum_i y_i, \qquad r(y) = c(y) - \ell(y), \qquad W = \mathsf{X} - \ell(Y).$$

Then $\mathbb{E}(W Y_i) = 0$. Sign symmetry eliminates cross terms in $\mathbb{E}r(Y)^2$; using the second, fourth and sixth moments gives

$$\begin{aligned} V_n := \mathbb{E}r(Y)^2 &= n\left(\frac{15}{9N(N+2)(N+4)} - \frac{2}{N(N+2)^2} + \frac{1}{N(N+2)^2}\right) \\ &= \frac{2n(N-1)}{3N(N+2)^2(N+4)}. \end{aligned} \tag{3}$$

Also $\mathbb{E}h(Y)^2 = 24/[N(N+2)]$. Consequently

$$D_n := \mathbb{E}W^2 = V_n + \frac{8}{N(N+2)}, \qquad n^2 D_n \longrightarrow \frac{26}{3}. \tag{4}$$

Write the canonical affine projection as $P_{K_n} g_n(x, y) = a_n x + b_n \cdot y$; its constant term vanishes by central symmetry and oddness. In coordinates (W, Y) the covariance of W with all Y_i is zero. The regression normal equations and (1) therefore yield

$$a_n = \frac{\mathbb{E}[W f_n(Y)]}{\mathbb{E}W^2} = \frac{\mathbb{E}[r(Y) f_n(Y)]}{D_n}. \tag{5}$$

We evaluate the numerator by a one-dimensional limit. Let $Z_n = \sqrt{n}\,Y_1$. For $i \neq j$, the integrand $(Y_i^3/3 - Y_i/(N+2))\varphi(\sqrt{n}Y_j)$ is odd in Y_i, so its expectation vanishes. All diagonal terms are equal. Hence

$$n^{3/2}\mathbb{E}[r(Y)f_n(Y)] = \mathbb{E}\left[\left(\frac{Z_n^3}{3} - \frac{n}{n+6}Z_n\right)\varphi(Z_n)\right]. \tag{6}$$

Now Z_n converges in distribution to a standard Gaussian Z. For example, represent the uniform spherical point by independent standard Gaussians (G_j) divided by their Euclidean norm. Then

$$Z_n \overset{\text{law}}{=} \frac{G_1}{\sqrt{n^{-1}\sum_{j=1}^{n+4} G_j^2}} \longrightarrow G_1$$

almost surely in this coupling by the law of large numbers. Moreover (2) gives

$$\mathbb{E}Z_n^8 = \frac{105n^4}{N(N+2)(N+4)(N+6)} \leq 105.$$

Since $|\varphi(t)| \leq |t|$, this eighth-moment bound gives uniform integrability of the fourth-degree expressions in (6). We obtain

$$n^{3/2}\mathbb{E}[r(Y)f_n(Y)] \longrightarrow \gamma := \mathbb{E}\left[\left(\frac{Z^3}{3} - Z\right)\varphi(Z)\right].$$

Gaussian integration by parts twice shows

$$\begin{aligned}\gamma &= \frac{1}{3}\mathbb{E}[(Z^2-1)\varphi'(Z)] \\ &= \frac{1}{3}\mathbb{E}[(Z^2-1)\min\{Z^2,1\}] \\ &= \frac{1}{3}\mathbb{E}[(1-Z^2)^2\mathbf{1}_{\{|Z|\leq 1\}}] > 0.\end{aligned}$$

In the last equality we subtracted the mean-zero variable Z^2-1. The integration by parts is valid for this C^1 function with linear growth and bounded derivative; explicitly, $\mathbb{E}[Z\varphi(Z)] = \mathbb{E}\varphi'(Z)$ and $\mathbb{E}[Z^3\varphi(Z)] = 2\mathbb{E}\varphi'(Z) + \mathbb{E}[Z^2\varphi'(Z)]$.

Equations (4)–(6) now give

$$\frac{a_n}{\sqrt{n}} \longrightarrow \frac{3\gamma}{26} = \frac{1}{26}\mathbb{E}[(1-Z^2)^2\mathbf{1}_{\{|Z|\leq 1\}}] = c_*.$$

Evaluating the projected affine function between 0 and e_0 yields $\mathrm{Lip}_{K_n}(P_{K_n}g_n) \geq |a_n|$.

Finally choose an invertible linear map L_n taking K_n to an isotropic body $\widetilde{K_n}$ of volume one. Apply the inverse square root of its positive definite covariance matrix, followed by a scalar dilation, to obtain such a map. Set $\widetilde{f_n} = g_n \circ L_n^{-1}$. Normalized uniform measures push forward under L_n, and affine maps are carried to affine maps, so

$$P_{\widetilde{K}_n}\widetilde{f}_n = (P_{K_n}g_n) \circ L_n^{-1}.$$

Also $\|L_n z\|_{\widetilde{K}_n} = \|z\|_{K_n}$, preserving both Lipschitz constants. On the normalized isotropic body this projection is exactly the formula in AIM 898. This proves the theorem. □

Corollary 1.2 (Arbitrarily small Banach–Mazur distortion)**.** *For every $D > 1$, the failure in Theorem 1.1 occurs on a family of normed spaces whose Banach–Mazur distances to Euclidean spaces are at most D.*

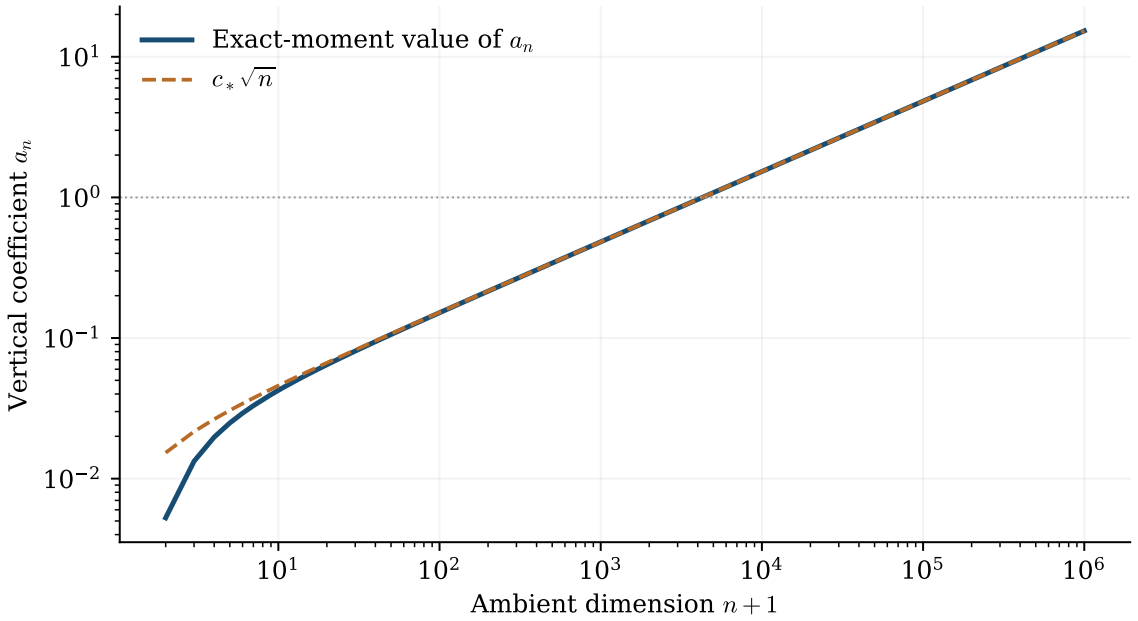


Figure 1: The vertical coefficient a_n, a lower bound for $\mathrm{Lip}_{K_n}(P_{K_n} g_n)$, evaluated from the exact beta-moment formula. The dashed curve is the proved asymptote $c_* \sqrt{n}$. The plotted evaluations illustrate the proof; no numerical estimate is needed for the unboundedness conclusion.

Proof. Fix $0 < \delta < 1/2$, and choose fixed parameters

$$M \geq \max\{1, (2\delta - \delta^2)^{-1/2}\}, \qquad 0 < \alpha \leq \min\{1/2, 3\delta\}.$$

Replace the preceding fiber functions by

$$h_M(y) = \min\{\sqrt{1 - \|y\|_2^2},\, M(1 - \|y\|_2^2)\}, \qquad c_\alpha(y) = \frac{\alpha}{3} \sum_i y_i^3,$$

and let $K_n^{M,\alpha}$ have fibers $|x - c_\alpha(y)| \leq h_M(y)$ over B_2^n. Both branches of $h_M \pm c_\alpha$ are concave: the Hessian of $\sqrt{1 - \|y\|_2^2}$ is at most $-I$, the other branch has Hessian $-2MI$, and $\|\nabla^2 c_\alpha(y)\| \leq 2\alpha \leq 1$. The minimum of two concave functions is concave. Thus these are again origin-symmetric convex bodies.

They satisfy the dimension-independent inclusions

$$(1 - \delta) B_2^{n+1} \subseteq K_n^{M,\alpha} \subseteq (1 + \delta) B_2^{n+1}. \tag{7}$$

For the outer inclusion, each fiber is contained in a vertical translate of the corresponding unit-ball fiber, with displacement $|c_\alpha(y)| \leq \alpha/3 \leq \delta$. For the inner inclusion, write $r = \|y\|_2 \leq 1 - \delta$. Our choice of M gives $h_M(y) = \sqrt{1 - r^2}$. Furthermore

$$\sqrt{1 - r^2} - \sqrt{(1 - \delta)^2 - r^2} = \frac{2\delta - \delta^2}{\sqrt{1 - r^2} + \sqrt{(1 - \delta)^2 - r^2}} \geq \delta.$$

This gap absorbs the same displacement $|c_\alpha(y)|$.

Use the identical explicit functions $g_n(x, y) = f_n(y)$ from the theorem; they remain intrinsically 1-Lipschitz. We give the measure comparison needed to retain the regression asymptotic. Introduce the reference body $\overline{K}_n^{M,\alpha}$ with the same centers c_α and untruncated half-width $M(1 - \|y\|_2^2)$, and let $\overline{\mu}_n$ be its normalized uniform measure. Its marginal $\|Y\|_2^2$ has the beta distribution with parameters $n/2$ and 2, exactly as in the main proof. The actual body is contained in this reference

body and coincides with it whenever $\|y\|_2^2 > 1 - M^{-2}$. Writing $a = n/2$ and $\rho^2 = 1 - M^{-2} < 1$, the probability of the exceptional region is bounded by

$$p_n := \overline{\mu}_n\{\|Y\|_2^2 \le \rho^2\} = a(a+1)\int_0^{\rho^2} s^{a-1}(1-s)\,ds \le (a+1)\rho^{2a}. \tag{8}$$

This decays exponentially in n. Normalized uniform measure μ_n on the actual body is $\overline{\mu}_n$ conditioned on that body. For every bounded function F, consequently,

$$\left|\int F\,d\mu_n - \int F\,d\overline{\mu}_n\right| \le \frac{2p_n}{1-p_n}\|F\|_\infty.$$

Here $|x| \le M + \alpha/3$, $\|y\|_2 \le 1$, and $|f_n(y)| \le 1$, uniformly in n. All scalar moments appearing in the regression normal equations therefore differ by $O_{M,\alpha}(p_n)$. To see explicitly that inversion introduces only polynomial losses, use coordinate symmetry to write

$$u = \mathbb{E}Y_1^2, \quad v = \mathbb{E}(\mathsf{X}Y_1), \quad t = \mathbb{E}(Y_1 f_n(Y)), \quad A = \mathbb{E}(\mathsf{X}f_n(Y)), \quad B = \mathbb{E}\mathsf{X}^2.$$

The vertical regression coefficient for either measure is

$$a_n = \frac{A - nvt/u}{B - nv^2/u}.$$

For the reference measure $u = 1/(n+4)$; for the actual measure $u = 1/(n+4) + O_{M,\alpha}(p_n)$, hence $u^{-1} = O(n)$. The two numerators and the two denominators in this formula differ by $O_{M,\alpha}(n^3 p_n)$, which is smaller than every fixed inverse power of n.

For the reference body, the original moment calculation applies with residual $\alpha r(Y)$ and conditional noise variance $M^2 h(Y)^2/3$. Its regression numerator is asymptotic to $\alpha\gamma n^{-3/2}$ and its denominator to $(2\alpha^2/3 + 8M^2)n^{-2}$. The preceding error bounds show that the actual coefficient satisfies the same limit:

$$\frac{a_n}{\sqrt{n}} \longrightarrow \frac{\alpha\gamma}{2\alpha^2/3 + 8M^2} > 0.$$

The vector e_0 still has intrinsic norm one, because $h_M(0) = 1$. Thus the projection norms are unbounded on the actual bodies. Their Banach–Mazur distances to Euclidean space are at most $(1+\delta)/(1-\delta)$ by (7). Choose δ small enough that this ratio is at most D, and apply the affine normalization from the theorem. The Banach–Mazur bound and the projection Lipschitz ratio both survive this normalization. □

Remark 1.3 (Distortion tending to one). The same family can have Banach–Mazur distance tending to one while its projection norms diverge. For sufficiently large n, take

$$\delta_n = n^{-1/8}, \qquad \alpha_n = 3\delta_n, \qquad M_n = (2\delta_n - \delta_n^2)^{-1/2}.$$

These parameters satisfy the preceding concavity and inclusion conditions. The inclusions give Banach–Mazur distance at most $(1+\delta_n)/(1-\delta_n) = 1 + O(n^{-1/8})$. Moreover $M_n \sim 2^{-1/2} n^{1/16}$ grows only polynomially, and the reference probability of the removed region is at most

$$(n/2+1)(1 - M_n^{-2})^{n/2} \le (n/2+1)\exp(-n\delta_n/2).$$

All bounds for the moments and the inversion of u in the preceding proof therefore still introduce only polynomial factors, which are negligible compared with this exponential decay. For the reference measure the numerator is $\alpha_n n^{-3/2}(\gamma + o(1))$, by exactly the same one-variable limit as before. Its denominator satisfies

$$\alpha_n^2 V_n + \frac{8M_n^2}{(n+4)(n+6)} \sim 8M_n^2 n^{-2} \sim 4n^{-15/8}.$$

The actual vertical coefficient consequently obeys

$$a_n \sim \frac{3\gamma}{4}\, n^{1/4} \longrightarrow \infty.$$

The isotropic normalization preserves both this lower bound for the projection Lipschitz constant and the Banach–Mazur distance.

References

[1] T. Hytönen and A. Naor, *Heat flow and quantitative differentiation*, J. Eur. Math. Soc. **21** (2019), 3415–3466. doi:10.4171/JEMS/906. Questions 10 and 19, and Theorem 9.